BD BATTLE PRO (ARENA X)

Master Product Requirements Document (PRD) & Technical Architecture Specification

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Platform Codename: bd-battle-pro / ARENA X
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1. Executive Summary & Market Opportunity

BD Battle Pro (ARENA X) is Bangladesh's first dedicated, autonomous, mobile-first esports operating system specifically designed for Free Fire 12-Squad competitive tournaments. Over 10,000 independent tournament organizers in Bangladesh run daily custom room matches. Currently, 99% of organizers manually operate through WhatsApp and Messenger, suffering from massive chat chaos, fake bKash/Nagad payment screenshots, recycled Transaction IDs (TrxID), room password leakage to non-payers, and tedious manual scoreboard counting.

BD Battle Pro solves every friction point by introducing automated SMS gateway payment verification, an autonomous 24/7 WhatsApp AI booking bot, phone-locked room credential release countdowns, and an automated AI scoreboard OCR engine with 0% gateway commission fees.

2. Core Problem vs. Architectural Solution Matrix

Traditional Organizer Bottlenecks	BD Battle Pro / ARENA X Automated Solution
Fake / Reused bKash & Nagad TrxIDs: Players edit screenshots or submit old TrxIDs to get free slots.	**SifrPay SMS Gateway APK:** Intercepts raw SMS from 16247/16167, performs cryptographic Firestore deduplication, rejecting duplicates instantly.
Chaotic WhatsApp Booking Volume: Organizers answer hundreds of repetitive DMs every day.	**Autonomous WhatsApp AI Bot (Baileys):** Greets players, dynamically presents active match schedules, validates match-specific fees, and books slots 24/7.
Custom Room Passwords Leaked: Password shared in group chats gets entered by non-paying outsiders.	**Phone-Locked Room Vault:** Credentials released strictly 15 minutes before match start ONLY to registered squad leader phone numbers.
Manual Points Table Math: Calculating kills and placements manually causes disputes and delays.	**AI Scoreboard Analyzer:** Google Gemini Vision OCR reads end-game Free Fire screenshots and calculates official FFWS points tables automatically.
Heavy SaaS Commission Fees: Payment gateways deduct 2.5% to 4% per transaction.	**0% Platform Commission:** Direct P2P Send Money to organizer's personal bKash/Nagad accounts.

3. System Technology Stack

- **Frontend & UX:** Next-gen HTML5, Vanilla ES2022 modules, Tailwind CSS, Facebook-grade dark UI (#18191a canvas, #242526 surfaces, #1877f2 primary accent), 100dvh dynamic viewport, touch-first mobile navigation.
- **Typography & Currency:** Hind Siliguri (Bengali) + Inter (Latin). Exclusively BDT (৳ / Taka).
- **Backend & Database:** Google Cloud Firebase Firestore with realtime snapshot listeners (`onSnapshot`).

- **Cloud Hosting:** Google Firebase Hosting (SSD CDN, global edge distribution, HTTP/2 & HTTP/3).
- **Autonomous WhatsApp Bot:** Node.js 20, Baileys Web multi-device socket, Render.com 24/7 cloud container with keepalive ping.
- **AI Core:** Google Gemini 2.0 Flash + 1.5 Flash dual-model quota balancing + Offline Banglish RegEx NLP parser.
- **Native Android App:** Kotlin, Gradle 8.13, Android SDK 34, NotificationListenerService, Room SQLite.

4. Module-by-Module Functional Architecture

Module 1: Public Showcase & Sales Landing Page (index.html)

- **Hero Section:** High-conversion headline, direct WhatsApp chat CTA (+8801791357521), and official email contact (sifrarchitect@gmail.com).
- **Live Interactive Demos:** One-tap access to live Player Portal (/player) and Admin Panel (/admin).
- **Clean Pricing Architecture:** Removed hardcoded numbers (৳4,999/৳9,999), replacing with Custom Starter and Enterprise Suite packaging.

Module 2: Player Experience Portal (player.html / /player)

- **Stacked Match Feed:** Carousel of daily matches with dynamic time, map, fee (৳BDT), prize pool, and status badges (Upcoming, Live, Finished).
- **12-Squad Slot Grid:** Realtime 12-slot visual grid showing Free Slots vs Booked Squads.
- **Instant Booking Modal:** Input Squad Name, Leader Name, WhatsApp Phone, Payment Method, and TrxID with 1-tap copy of organizer's bKash/Nagad number.
- **Room ID & Password Vault:** Secured by phone verification; decrypts room credentials only for confirmed players 15 minutes before game start.

Module 3: Organizer Admin Headquarters (admin.html / /admin)

- **Security Gate:** Passcode-protected access (default: ARENA2026).
- **Financial KPI Ledger:** Realtime calculation of Confirmed Inflow, Prize Pool Committed, and Net Profit in BDT.
- **Match Manager:** Create, edit, and delete matches with maps (Bermuda, Purgatory, Alpine, Kalahari, Nexterra), times, and fees.
- **12-Slot Controller:** Kick/reset slots or manually lock slots for VIP squads.
- **Room Credentials Switch:** Store Room ID & Password and toggle visibility to players.
- **SifrPay APK Download:** 1-click download of the clean Android SMS Gateway APK (.zip package).
- **AI Scoreboard OCR:** Upload end-game screenshot to auto-generate FFWS points table.

Module 4: Autonomous WhatsApp AI Bot (whatsapp-bot)

- **Mobile Pairing Code (server.js):** 8-digit mobile pairing code interface; organizers connect WhatsApp directly from mobile without scanning QR codes.
- **Dynamic Tournament & Fee Selector:** Bot greets players, displays live match schedule with exact fees, and requires players to choose a match before asking for payment.
- **Dual-Model Quota Balancing:** 50/50 round-robin rotation between Gemini 2.0 Flash and 1.5 Flash to eliminate rate limits.
- **Offline RegEx Fallback:** Zero-dependency Banglish NLP parser that guarantees 100% booking uptime even if Gemini API quota is exhausted.

Module 5: SifrPay Android SMS Gateway APK

- **Zero-Setup Experience:** Completely removed Merchant ID and Secret Key setup dialogs; app opens directly to listening dashboard.
- **Automated Interception:** Reads incoming SMS notifications from bKash (16247) and Nagad (16167) in background, syncing verified TrxIDs to Firestore.

Module 6: Production Build & Anti-Theft Guard (protect-build.js)

- Source code preserved in `src_backup/`; compiled HTML obfuscates scripts and disables desktop DevTools inspection while keeping mobile 100% lightweight.

5. Current Status & Deliverables Summary

Milestone / Feature	Status	Notes
Facebook-Grade Dark UI (Player & Admin)	COMPLETED	Live on Firebase Hosting
Realtime Firestore Slot Sync & Phone Vault	COMPLETED	12-squad atomic locking
SifrPay APK (Zero-Setup Clean Build)	COMPLETED	v1.3 released; no merchant dialog
APK Download from Firebase CDN (.zip)	COMPLETED	Verified 200 OK download
WhatsApp Bot 24/7 Cloud Host & Pairing Code	COMPLETED	Render Docker container ready
Dynamic Tournament & Fee Selector Bot	COMPLETED	Asks match first, gets exact fee
Dual-Model Gemini Rotation + Offline Fallback	COMPLETED	50/50 quota split + RegEx parser
Anti-Theft Obfuscation (protect-build.js)	COMPLETED	Protected production build
Official Contact & Pricing Update	COMPLETED	sifrarchitect@gmail.com updated

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